Mother Bats' Hearts Speed Up When They Hear Their Own Pups

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A bat **colony** can hold hundreds or even thousands of bats crowded together. So how does a mother bat find her own baby? A new study suggests she listens closely, and her heart reacts.

Biologists at Doshisha University in Japan noticed something in their lab **colony** of Japanese house bats. Whenever a young bat, called a **pup**, wandered too far from its resting spot, only its own mother rushed to help.

Pups make special calls when they are separated, a bit like a human baby crying. Each **pup**'s call has its own sound. "What is happening inside the mother when she hears these calls?" wondered biologist Kazuki Yoshino-Hashizawa.

To find out, the team recorded the calls of young female **pups** when they were two days old. These bats grow up fast. They began using **echolocation**, the sound-based way bats find their way, at about eight days old, and flew like adults after about a month. The researchers recorded those grown-up calls too.

About 40 days after the **pups** were born, the scientists played the recordings to four mother bats while measuring their heartbeats. The mothers' hearts beat faster when they heard their own **pups**, but not other mothers' **pups**. Surprisingly, the biggest change came from the calls of the older **pups**.

The mothers' hearts also sped up when they heard an unfamiliar adult male bat. The scientists say a faster heartbeat cannot show whether a bat feels caring or alarmed.

Still, the researchers think a **pup**'s calls quickly get a mother ready to care for her baby. The study was published in the **journal** Frontiers in Ethology.

Word Treasure

colony -- A large group of animals of the same kind that live together.
pup -- A baby bat.
echolocation -- A way bats find their way by making sounds and listening to the echoes.
biologists -- Scientists who study living things.
journal -- A magazine where scientists publish their research.

Background you need

Bats are mammals, so mothers feed and care for their babies, called pups, after they are born.

Many bats live in crowded colonies and rely on sound more than sight, using echolocation to find their way.

Scientists often measure heart rate to see how an animal's body reacts to something, though a faster heartbeat can mean excitement or alarm.

Recognizing a baby's unique call helps mothers in large groups find and protect their own young.

Break it down

LEAD: Mother bats' hearts beat faster when they hear their own pups
QUESTION: How do mothers find their pups in crowded colonies?
OBSERVATION: At Doshisha University, only a pup's own mother rescued it when it wandered off
METHOD: Recorded Japanese house bat pups' calls and played them to four mothers while measuring heart rates
+- RESULT: Hearts sped up for their own pups, not others
+- SURPRISE: Biggest change came from older pups' calls
CAUTION: Hearts also sped up for an unfamiliar adult male, so feelings cannot be read from heart rate alone
PUBLISHED IN: Frontiers in Ethology
OPEN QUESTION: Why did older pups' calls cause the strongest reaction?

Why it matters

The study shows how animals can recognize family members by sound alone, and gives scientists a new way to measure how animals react to the world around them.

Test what you remember

1. Which university did the scientists who did this study work at?

- (A) University of Oxford
- (B) Doshisha University
- (C) University of Turin
- (D) University of Cambridge

2. What did the scientists notice when a pup wandered too far?

- (A) All the bats rushed over
- (B) Only its own mother raced to help
- (C) No bat noticed
- (D) The pup flew back by itself

3. What are isolation calls like?

- (A) A dog's bark
- (B) The cries of a human baby
- (C) A cat's purr
- (D) A bird's song at dawn

4. How many mother bats were tested?

- (A) One
- (B) Forty
- (C) Four hundred
- (D) Four

5. Which recordings made the mothers' heart rates rise the most?

- (A) Music
- (B) Silence
- (C) Calls of other mothers' pups
- (D) Calls of the older, independent pups

6. Why can't heart rate alone show how the mothers felt?

- (A) The bats were asleep
- (B) It cannot tell caring feelings from alarm
- (C) The machine was broken
- (D) Bats have no heartbeat

Answer key (try the quiz first!): 1: B 2: B 3: B 4: D 5: D 6: B

Questions to think about

1. Kazuki Yoshino-Hashizawa, biologist: Wondered what happens inside a mother bat when she hears her pup's calls and suggests the calls prime her to care for it.

MY THOUGHTS (at least 20 words):